

File permissions in Linux

Project description

The research team wants to ensure users on this team are authorized with the appropriate permissions to help keep the system secure.

I am in charge of examining existing permissions on the file system. I need to determine if the permissions match the authorization that should be given. If they do not match, I am to modify the permissions to authorize the appropriate users and remove any unauthorized access.

Check file and directory details

```
researcher2@e16d0e29bfd9:~$ ls
projects
researcher2@e16d0e29bfd9:~$ cd projects
researcher2@e16d0e29bfd9:~/projects$ ls
drafts project_k.txt project_m.txt project_r.txt project_t.txt
researcher2@e16d0e29bfd9:~/projects$ ls -la
.  .. .project_x.txt drafts project_k.txt project_m.txt project_r.txt project_t.txt
researcher2@e16d0e29bfd9:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:17 ..
-rw--w--- 1 researcher2 research_team  46 Feb 21 07:13 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$
```

I have used `ls` to display the available directories or files and navigated to the projects directory using `cd projects`. To display the permission of all files including hidden files, the `ls` command has been used with the `-l` and `-a` flags for long listing format permissions and all files respectively.

Describe the permissions string

The permissions string, output by the `ls -al` command is a 10-character string that represents file type and access permissions for files and directories present in the working directory. Using the permissions for `project_k.txt` `-rw-rw-rw-` as an example, the first character of the string `-`, tells us that this is a file, whereas `d` would be used for a directory.

The remaining 9 characters are broken up into triads where each represents the user (`-rw`), group (`rw-`) and others (`rw-`). Each triad follows the order `rwX` and a `-` indicates that a specific permission is not granted.

Therefore, we can see that `-rw-rw-rw-` means that, in the file `project_k.txt`, the user, group and others have read and write permissions only and none have execute permissions.

Change file permissions

The organization does not allow others to have write access to any files. The file `project_k.txt` has write permissions. This needs to be updated.

```
researcher2@e16d0e29bfd9:~/projects$ chmod o-w project_k.txt
researcher2@e16d0e29bfd9:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$
```

The `chmod` command is used with its arguments, `o-w` and the target file name, `project_k.txt`. In the 10-character string, each triad is represented by a character, `u` for user, `g` for group and `o` for other. The `-` in `o-w` shows that we are removing a permission and the `w` is the permission being removed. This is reflected in the updated `ls -l` command output which shows that the write permission for the file is removed.

Change file permissions on a hidden file

The research team has archived `.project_x.txt`, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file.

```

researcher2@e16d0e29bfd9:~/projects$ chmod ug-w .project_x.txt
researcher2@e16d0e29bfd9:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:17 ..
-r----- 1 researcher2 research_team  46 Feb 21 07:13 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$

```

```

researcher2@e16d0e29bfd9:~/projects$ chmod g+r .project_x.txt
researcher2@e16d0e29bfd9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:13 .
drwxr-xr-x 3 researcher2 research_team 4096 Feb 21 07:17 ..
-r--r----- 1 researcher2 research_team  46 Feb 21 07:13 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$

```

For hidden files, the target file argument is prefixed with a `.` and the `ls -la` flag needs to be used to view the changes.

`chmod` allows for multiple groups to be updated at once as shown in the `ug-w` argument that removes the write permission for user and group in one command. Then, the `g+r` is used to give the group, `g`, the read permission as specified by the research team. The `+` and `-` tags are an intuitive way to give and remove permissions.

Change directory permissions

The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents.

```
researcher2@e16d0e29bfd9:~/projects$ chmod g-x drafts/
researcher2@e16d0e29bfd9:~/projects$ ls -l
total 20
drwx----- 2 researcher2 research_team 4096 Feb 21 07:13 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_k.txt
-rw----- 1 researcher2 research_team  46 Feb 21 07:13 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Feb 21 07:13 project_t.txt
researcher2@e16d0e29bfd9:~/projects$
```

The chmod command can be used in the same way for directories as for files.

Summary

Ensuring users on this team are authorized with the appropriate permissions is essential to help keep the system secure. Performing regular audits is important to prevent permission creep.

The existing permissions on the file system were audited using `ls -al`. Utilised `chmod` to change the permissions for the files and directories based on the specifications by the research team. An in-depth understanding of bash commands and the 10-character symbolic notation permission string was required to verify changes.

These tasks enforce the principle of least privilege, where authorization for appropriate users is allowed and any unauthorized entry be strictly restricted. This ensures members on the organisation have only what they need to do their job and nothing more. This is especially important to maintain the integrity of archived hidden files on the organisation network.